PROJECT PROPOSAL

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1. Project Title

Sentiment Analysis of myBCA Mobile Banking App Reviews.

2. Problem Statement

The myBCA mobile banking application receives thousands of user reviews on the Google Play Store daily. Reading all reviews manually is inefficient, causing important feedback about bugs or service quality to be missed. This project builds an automatic sentiment classification system using deep learning to classify each review as Positive, Neutral, or Negative it's helping the myBCA team quickly identify user issues without manually reading every review. Understanding user sentiment is critical for improving mobile banking services and maintaining customer trust.

3. Dataset

Dataset Name: myBCA Google Play Store Reviews Dataset

Source: Collected via web scraping from the myBCA page on Google Play Store (https://play.google.com/store/apps/details?id=com.bca.mybca) using the google-play-scraper Python library.

Size: Approximately 5,000 user reviews in Bahasa Indonesia.

Input: Review text written by users. Target Labels: Positive (4–5 stars), Neutral (3 stars), Negative (1–2 stars).

Format: CSV file with columns: review, rating, sentiment.

4. Planned Method

Baseline Model: Logistic Regression with TF-IDF text representation, serves as a benchmark to compare against the deep learning model.

Deep Learning Model: LSTM (Long Short-Term Memory), chosen because it is designed for sequential data and can understand the context of words in a sentence, making it well-suited for sentiment analysis.

Loss Function: Categorical Cross-Entropy (standard for 3-class classification).

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score (weighted), and Confusion Matrix.

Data Split: Training 70% (~3,500 reviews), Validation 15% (~750 reviews), Test 15% (~750 reviews).

5. Expected Challenges

- **Class Imbalance**: Positive reviews may dominate the dataset. Will be handled using class weighting during training.
- **Informal Language**: Reviews contain typos, slang, and mixed Indonesian-English text, requiring careful preprocessing.
- **Ambiguous Neutral Reviews**: 3-star reviews often contain mixed opinions, making them harder to classify correctly.
- **Limited Compute**: Google Colab free tier has GPU limitations; model checkpoints will be saved regularly to avoid data loss.

6. Weekly Plan

Week	Planned Work	Expected Output
Week 1	Dataset scraping from Google Play Store, EDA, GitHub repo setup	~5,000 reviews CSV, EDA notebook, README, Week 1 report
Week 2	Text preprocessing, sentiment labeling, train/val/test split, baseline model (Logistic Regression + TF-IDF)	Cleaned dataset, preprocessing script, baseline results, Week 2 report
Week 3	LSTM model training, hyperparameter tuning, evaluation with Accuracy, F1-Score, Confusion Matrix	Trained LSTM, evaluation results, baseline vs LSTM comparison, Week 3 report
Week 4	Model improvement, error analysis, final report writing, presentation preparation	Final model, final report, presentation slides, Week 4 report